# Mekong Quality Assurance Advisory

### Keeping production true to what you approved

How the Product Spec, Packaging Specification and Verification Protocol fit together, and who owns each.

You approved a product. The risk in co-packing is that later production runs slowly stop matching it: a little darker, a little thinner, a different cap, a label with a printing error. Nobody changed the recipe on purpose, and nobody can prove what the standard was, because it was never written down tightly enough to check. I call that drift.

I prevent it with three working documents. The Product Spec fixes what your approved product is, as measurable ranges. The Packaging Specification fixes what the pack, label and pallet must be. The Verification Protocol is how I check each production run against both. Your counsel attaches them to the agreement with your co-packer, so they are enforceable.

I work from your approved finished product, not from your recipe. The recipe stays yours.

**Naming.** Your own paperwork may call these documents an annex or exhibit. What matters is that the content exists.

**Scope:** Quality and consistency verification against your specifications.

## The four documents at a glance

| Document | What it is | Drift it prevents |
|---|---|---|
| **Master Quality Agreement** | The signed agreement between you and the co-packer that the other three documents attach to, written by your counsel: consequences of failure, change control, per-lot test data, retained samples, traceability, recalls. | Nothing makes the plant answerable when a lot fails, and changes happen without sign-off. |
| **Product Spec** | The food: color, °Brix, pH, water activity, salinity, alcohol by volume (ABV) where applicable, consistency, fill, visual, and taste and smell, as measurable ranges built from your approved product. | The product slowly moving away from the one you approved. |
| **Packaging Specification** | The pack: seal and closure, label and printing, ingredient list and allergen statement as printed, barcode, lot and date code, carton, pallet. | Quiet changes in materials and pallets, and printing errors that reach the shelf. |
| **Verification Protocol** | How each production run is checked against the other two: what I measure on site, sampling plans, defect classes, what happens when a lot fails. | The first two documents sitting on paper with nobody checking them. |

## The big picture

Teal: I write it or carry it out. Amber: you own or approve it. Gray: the plant or your regulatory advisor owns it.

**MASTER QUALITY AGREEMENT** *Owned by you and your counsel*

- Makes the three documents below binding. Holds consequences of failure (reject, retest, hold), change control, per-lot test data, retained samples, traceability, recalls.

*Drift prevented: nothing makes the plant answerable when a lot fails.*

**PRODUCT SPEC** *I write it; you approve*

**The food: quality and consistency**

- Color, °Brix, pH, water activity, salinity and ABV (where applicable): lab baselines, set as ranges
- Bostwick consistency, fill, visual (specks, gloss, separation)
- Taste and smell against a retained reference sample and a short flavor profile
- Named off-notes, zero tolerance: burnt, metallic, rancid, fermented
- Each attribute measured by a method defined for your product
- Initial ranges tighten by an agreed rule after the first lots

**Built from your approved product, not your recipe.**

*Drift prevented: the product slowly moving away from the original.*

**PACKAGING SPECIFICATION** *You own the standard; I record it*

**The pack: bottle, closure, label, carton, pallet**

- Path 1: your packaging consultant writes it
- Path 2: the plant states its specs, I record them, you approve
- Pack and seal, cap torque, cartons, pallet pattern and wrap
- Print quality and printing errors; barcode; lot and date code
- Ingredient list and allergen statement as printed, matched to your approved artwork (legal compliance is for your advisor)
- Light document check; not a structural engineering review

*Drift prevented: quiet changes in materials, print and pallets.*

**VERIFICATION PROTOCOL** *I write it and carry it out*

**ON SITE, with my own instruments**

- Bostwick consistometer, refractometer (°Brix), pH meter, water activity meter; salinity and ABV (where applicable) per the product's defined method
- Color: per the product's defined method, after a calibration check
- Taste and smell against the reference; blind panel for quality calls
- Fill, seals, labels, print, codes; cartons, pallets, foreign matter
- AQL sampling (ISO 2859-1; US equivalent ANSI/ASQ Z1.4); critical, major and minor defects

**ACCREDITED LAB: BASELINES AND DISPUTES**

- Baseline readings when the Product Spec is set: color, water activity, pH, °Brix, salinity, ABV (where applicable)
- Disputed readings: retained split sample to a pre-agreed lab
- I collect, seal, ship and track every sample; the lab tests; results are reported against your specification
- One tamper-sealed sample kept from each inspection

**Inspection report to you, with readings and photos**

**You decide: release, hold, rework or reject**

**OWNED BY OTHERS: the plant and your regulatory advisor**

- HACCP, process validation, certification, destination-market label legality.

## Who owns what

| Component | Prepared by | Approved by | My role | Where my role ends |
|---|---|---|---|---|
| **Recipe and formulation** | You | You | I work from the finished product you approve, not from the recipe. | Writing, changing or reviewing your recipe. |
| **Master Quality Agreement** | Your counsel | You and the co-packer | I supply the three documents to attach, and advise on what the agreement should require. | Drafting legal terms or giving legal advice. |
| **Product Spec** | Me | You (the co-packer acknowledges it) | I set the ranges from lab baseline readings, define the measurement method for each attribute, and set the taste, smell and appearance reference. | Anything outside quality and consistency. |
| **Packaging Specification** | Path 1: your packaging consultant. Path 2: the co-packer states its specs. | You | Path 1: I turn it into inspection checks. Path 2: I request the plant's specs, record them, and run a light document check. | Engineering or designing the packaging, or certifying it will survive transit. |
| **Verification Protocol** | Me | You | I write it, carry out the inspections, and report the results. | Releasing or rejecting the lot. You decide. |
| **Food safety, certification and label legality** | The co-packer; your regulatory advisor | Certification body or regulator, where applicable | I check that what is printed matches the artwork you approved. | Owned by the plant and your regulatory advisor. |

## Product Spec: what goes in it

The baseline comes from your approved product. For color, water activity, pH, °Brix, salinity and, where applicable, ABV, an accredited lab takes every baseline reading, using the method defined for that product, so the standard rests on traceable numbers. I set the consistency and sensory baselines myself. Each parameter has a target value and a range, and each is measured by a method defined for your product and written into the specification. Where a Codex Alimentarius standard exists for your product category, I reference it when defining what the product is.

Initial ranges are time-boxed. For the first commercial lots (for example, the first three, or the first run if you produce less often), the ranges are initial ranges: they are outer limits and never loosen. After those lots they tighten by a method agreed in advance, such as recalculating from the first lots' data with a maximum width, and the tightened ranges take effect automatically, with no renegotiation. The target value stays fixed, and my reports track where each lot lands inside its range, so a plant drifting toward the loose edge is flagged even while lots still pass. Your counsel writes the automatic-tightening clause into the agreement.

The specification covers color, °Brix, pH, water activity, salinity, ABV (where applicable), consistency, fill and visual, and taste and smell. Taste and smell is defined by a retained sealed reference sample and a short flavor profile written with you (aroma, main flavor notes, balance, texture and finish), using the sensory terms defined in ISO 5492, plus a zero-tolerance list of named off-notes: burnt or scorched, metallic, rancid, fermented, bitter. How each attribute is checked, and with what, is in one table under the Verification Protocol.

### What sits outside the Product Spec, and who owns it

| Item | Where it lives | Owner |
|---|---|---|
| Allergens and cross-contact | The plant. The allergen statement as printed is checked against your approved artwork through the Packaging Specification. | Co-packer; you for the label |
| Contaminants (heavy metals, mycotoxins, pesticide residues) | Your regulatory requirements. | Your regulatory advisor |
| Microbiology | The plant controls it. | Co-packer |
| Additives, colorants, destination-market labeling rules | Your regulatory compliance. | You |
| Traceability and lot coding | Lot code format in the Packaging Specification; traceability duties in the agreement. | You and the co-packer |
| Per-lot test data, retained samples, change control, recalls, consequences | Master Quality Agreement. | Your counsel |

## Packaging Specification: two ways to build it

| | Path 1: you have a packaging consultant | Path 2: you use the co-packer's packaging |
|---|---|---|
| **Who defines the standard** | Your consultant writes the Packaging Specification. | The plant states what it can do; you agree to it. |
| **What I do** | I turn the specification into inspection checks. | I request the plant's packaging specs (seal, cap torque, carton strength, pallet pattern, artwork, barcode), record them as the Packaging Specification, and run a light document check, not an engineering review: food-contact materials, whether supplier test data exists for the carton and pallet, and whether the shipping route calls for a packaging consultant or a transit test. |
| **Your role** | You approve the consultant's specification. | You approve the recorded specification before it becomes the standard. |
| **What gets inspected** | The consultant's standard. | What the plant says it can do and you agreed to. |
| **What the record is** | The consultant's engineering, which I inspect against. | A baseline description of what the plant uses. It is not a structural load assessment, and who carries the risk of packaging that fails in transit is decided in the agreement. |
| **What I do not do** | Engineer or redesign the packaging. | Engineer or redesign the packaging. |

**Contents:** primary pack and seal; closure and cap torque; label and artwork, including print quality, smearing, misprints and color against the approved artwork; ingredient list and allergen statement as printed; net contents and claims as approved; barcode; lot and date code; master carton; pallet pattern and wrap.

I check that what is printed matches the artwork you approved. Whether that label is legally compliant is for your regulatory advisor.

In Path 2 the record describes what the plant uses. For a long or rough shipping route, I recommend a packaging consultant or a transit test, and your counsel should decide in the agreement who carries the risk if packaging fails in transit.

## Verification Protocol: what I check and how

The lab takes the baseline readings once, when the Product Spec is set. At each inspection, my on-site readings are checked against those baseline ranges. This one table shows each check, where its baseline comes from, and how it is done.

| Check | Baseline | Where | Method | Notes |
|---|---|---|---|---|
| **Color** | Accredited lab readings on your approved product; ranges set from the results. | On site | Per the product's defined method, after a calibration check. | Color is locked in as a number, so it can't be argued after the fact. If the plant has no suitable instrument, a retained sample goes to the accredited lab. |
| **°Brix** | Accredited lab baseline readings. | On site | Refractometer. | |
| **pH** | Accredited lab baseline readings. | On site | pH meter. | |
| **Water activity** | Accredited lab baseline readings. | On site | Water activity meter. | |
| **Salinity** | Accredited lab baseline readings. | On site | Per the product's defined method. | |
| **Alcohol by volume (ABV)** | Accredited lab baseline readings, where applicable to your product. | On site | Per the product's defined method. | |
| **Consistency** | My baseline readings on your approved product. | On site | Bostwick consistometer. | Per the product's defined method. |
| **Fill, seals, leaks and visual** | Your declared fill and the approved product. | On site | Fill check; visual and handling checks for specks, gloss, separation, seals and leaks. | |
| **Taste and smell** | Retained sealed reference sample, a short flavor profile written with you, and named off-notes with zero tolerance. | On site | Side by side with the reference and profile. Quality calls by blind panel. | I smell first. Off-notes such as rancid, fermented, moldy or chemical are reported at once and not put to a panel. If the panel can't agree, you decide. |
| **Labels, printing, barcode, lot and date codes** | Approved artwork in the Packaging Specification. | On site | Compared with the approved artwork; ingredient list and allergen statement read against it; barcode scanned. | Printing errors are recorded with photos. |
| **Cartons and pallets** | Packaging Specification. | On site | Against the Packaging Specification. | |
| **Visible foreign matter** | | On site | Visual. | |

My instrument calibration records are kept and available to you, so my readings are defensible. I also check any instrument I don't own before I accept its readings.

## If a reading is disputed

The first step is to measure the same sample side by side with the plant and compare calibration records. If it is still disputed, a retained split sample goes to a pre-agreed accredited lab, which measures it against the recorded numbers using the method named in the agreement. The binding terms, who pays for the lab, and the timing belong in the agreement your counsel writes.

**Retained samples.** The plant keeps sealed retains of each lot, and the number and the time they are kept are set in the agreement, scaled to the pack size. At each inspection I also take one tamper-sealed sample, log it and hold it, so a dispute sample never rests only on the plant's own storage.

## How a taste or smell call is made

Each run is compared with the retained reference and the flavor profile written with you: aroma, main flavor notes, balance, texture and finish. A difference from the profile is reported. Off-notes are handled as below.

Off-notes such as rancid, fermented, moldy, chemical or machine oil are not put to a vote. I report them immediately and hold the lot.

Quality calls (burnt or scorched, metallic, bitter, clearly off the reference) are made by a blind side-by-side comparison with the retained reference, by at least two tasters including the plant's QA person. A triangle test, where tasters pick the odd sample out of three, shows whether the production sample is reliably different from the reference. A quality hold never rests on my palate alone. If the panel can't agree, I give you the findings and the retained sample, and you decide.

## Defect classes

| Class | Examples | What happens |
|---|---|---|
| **Critical** | Visible foreign matter; leaks, seal failures or swollen packs; wrong product or wrong pack; missing or wrong allergen statement; off-notes such as rancid, fermented, moldy, chemical or machine oil. | I stop and report to you immediately. Lot on hold. |
| **Major** | °Brix, pH, water activity, salinity, ABV, color or consistency out of range; quality off-notes confirmed by the blind panel (burnt or scorched, metallic, bitter, clearly off the reference); fill out of range; printing errors or wrong claims on the label; wrong or unscannable barcode; unreadable lot or date code; crushed cartons; unstable pallet. | Counts against the AQL. A failed lot is held or rejected at your decision. |
| **Minor** | Slight label tilt; small cosmetic print flaws; smudged but legible date code; scuffed master cartons. | Counts against the AQL. Noted in the report. |

Sampling uses AQL under ISO 2859-1 (US equivalent ANSI/ASQ Z1.4), with limits set by lot size and product risk.

## Responsibilities outside my work

These areas belong to the plant and to your regulatory advisor.

| Item | Owner |
|---|---|
| Hazard analysis and process controls (HACCP) | Co-packer |
| Process validation | Co-packer and its process authority |
| Microbiological control | Co-packer |
| Certification | Co-packer and the certification body |
| Destination-market regulatory compliance and label legality | Your regulatory advisor |

### How an engagement runs

**1. Baseline.** You provide your approved product. An accredited lab takes baseline readings for color, water activity, pH, °Brix, salinity and, where applicable, ABV. I take the consistency baseline, write the flavor profile with you, retain the taste and smell reference, set the ranges, and define the measurement method for each attribute. This produces the Product Spec.

**2. Packaging.** Path 1 (your consultant writes it) or Path 2 (the plant states its specs and I record them). You approve. This produces the Packaging Specification.

**3. Protocol.** I write the Verification Protocol from the first two documents: instruments, on-site checks, AQL sampling, defect classes.

**4. Agreement.** You and your counsel attach the three documents to the Master Quality Agreement.

**5. Inspect.** I check each production run, scheduled or unannounced, and report with readings and photos. You decide to release, hold, rework or reject.

**6. Review drift.** After the first lots the ranges tighten by the method agreed in advance. I also update the documents through change control.